

PART A

3

1)

- a) Statement 1 $\rightarrow$ True ✓
Statement 2 $\rightarrow$ False ✓

Answer: a) ONLY Statement 1 is TRUE

~~b)~~

- b) Answer: (a) $\mathbb{R}_{57}$ is NOT an algebraic field, because it is not prime ✓

$\{2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37\}$

133 ✓

- c) (b) $S_1 = \left\{ \begin{pmatrix} 2 \\ 3 \end{pmatrix} \right\}$ not a vector space
as $\begin{pmatrix} 2 \\ 3 \end{pmatrix} + \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 4 \\ 6 \end{pmatrix} \notin S_1$

- d) Statement 1 $\rightarrow$ True (Even for zero vector subspace) ✓
Statement 2 $\rightarrow$ True (According to the definition of a basis, yes) ✓

Answer: ~~(a) Statement~~

(b) Both statements are TRUE ✓

- e) (c) The set is a vector space over a field of real numbers. ✓

a $\rightarrow$ T
b $\rightarrow$ T
d $\rightarrow$ T

10

PART B

CIVIL +

2010

4

- 1) Taking $v_1, v_2, v_3, v_4 \in V$ such that the set of these vectors are linearly independent.

This implies that no vector from this set can be expressed in terms of one or more vectors from the set, other than itself.

Since V has 4 linearly independent vectors, then each of these vectors must have at least four components

thus let $v_1 = [a \ 0 \ 0 \ 0 \ \dots]^T$

$$v_2 = [0 \ b \ 0 \ 0 \ \dots]^T$$

$$v_3 = [0 \ 0 \ c \ 0 \ \dots]^T$$

$$v_4 = [0 \ 0 \ 0 \ d \ \dots]^T$$

~~each that they contribute to index~~

let us form the vectors

$$v_1 + v_2 = [a \ b \ 0 \ 0 \ \dots]^T$$

$$v_1 + v_2 + v_3 = [a \ b \ c \ 0 \ \dots]^T$$

$$v_1 + v_2 + v_3 + v_4 = [a \ b \ c \ d \ \dots]^T$$

let us check for linear independence

$$\begin{bmatrix} a & 0 & 0 & 0 & \dots \\ a & b & 0 & 0 & \dots \\ a & b & c & 0 & \dots \\ a & b & c & d & \dots \end{bmatrix}$$

- $R_4 \rightarrow R_4 - R_3$
 - $R_3 \rightarrow R_3 - R_2$
 - $R_2 \rightarrow R_2 - R_1$
- } elementary row operations

$\Rightarrow$ We get
$$\begin{bmatrix} a & 0 & 0 & 0 \\ 0 & b & 0 & 0 \\ 0 & 0 & c & 0 \\ 0 & 0 & 0 & d \end{bmatrix}$$

- 6) which is the same as the earlier matrix.
- There are no zero rows in this row echelon matrix, ~~etc~~
 - all rows are linearly independent
- $\therefore v_1, v_1+v_2, v_1+v_2+v_3, v_1+v_2+v_3+v_4$ are also linearly independent

3) Statement 1: The set containing a single vector is always a linearly dependent set.

This statement can be REJECTED.

- Assume a non zero vector $\begin{pmatrix} a \\ b \end{pmatrix}$ where $a, b \in \mathbb{R}$ and $a \neq 0, b \neq 0$

In this single vector set, for the set to be independent, for $c \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$, c must be only and only 0

- c can take only the scalar value of 0.
- Hence this set of vectors is linearly independent.

Homener for a set containing only the zero vector, such as $\left\{\begin{pmatrix} 0 \\ 0 \end{pmatrix}\right\}$

6

For $c \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ to occur, c can take infinitely many values.

Hence this set is NOT linearly independent.

In conclusion, set containing a single vector is ALWAYS linearly independent if the single vector is NOT a zero vector.

Statement 2: The set containing a single vector is ALWAYS a basis for some vector ~~spa~~ subspace.

This statement is REJECTED.

The definition of basis is that it is a minimal, linearly independent spanning set.

This disqualifies the set containing only the zero vector, because the single vector set of the zero vector is not linearly independent as seen above.

zero vector single set is NOT even a basis for the zero vector vector space, whose basis is actually the null set.

In conclusion, the set containing a NON-ZERO single vector is a basis for some subspace.

Statement 3: Every vector space contains at least one subspace.

This statement is ASSERTED, because every vector space has at least either/and the entire vector space or/and the zero vector to be subspaces of the vector space.

4) a) In this statement, we will assume that non-zero subspace means a subspace that contains zero vector, but is not ONLY the zero vector, i.e. it represents more vectors than just the zero vector. (We know that a vector subspace cannot be a subspace without the zero vector).

(TRUE)

Yes, we can say that every non-zero subspace of $\mathbb{R}^5$ will have infinitely many bases. However, we must make a distinction between the dimension of the basis of this subspace, which is finite and the infinite combination of bases which you could use to represent this subspace.

For example, the entire subspace of $\mathbb{R}^5$ could be represented by bases

$$\{ (1, 0, 0, 0, 0), (0, 1, 0, 0, 0), (0, 0, 1, 0, 0), (0, 0, 0, 1, 0), (0, 0, 0, 0, 1) \}$$

OR

$$\{ (2, 0, 0, 0, 0), (0, 2, 0, 0, 0), (0, 0, 2, 0, 0), (0, 0, 0, 2, 0), (0, 0, 0, 0, 2) \}$$

OR... and so on

b) FALSE.

The set of columns of ANY 3×3 matrix cannot form a basis for $\mathbb{R}^3$.

eg: If we take $\begin{bmatrix} 1 & 2 & 0 \\ 1 & 2 & 0 \\ 1 & 2 & 0 \end{bmatrix}$ as the matrix,

we see that $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ is the only linearly independent vector, and it alone cannot form a basis for the entirety of $\mathbb{R}^3$. ✓

only
A matrix whose column vectors are linearly independent can form a basis for $\mathbb{R}^3$.

c) FALSE.

The rank of matrix A is the number of non-zero rows in the row-echelon form of the matrix A, not the matrix A itself. ✓

d) FALSE.

Let us understand using a smaller vector space with vectors having lesser components than $\mathbb{R}^{27}$, such as $\mathbb{R}^2$.

$\mathbb{R}^2$ contains the subspaces of zero vector subspace and entire vector space $\mathbb{R}^2$ itself. On top of that, multiple subspaces belonging to $\mathbb{R}^2$ can be defined using

$$S_X = \left\{ \begin{pmatrix} x \\ \alpha x \end{pmatrix}, x \in \mathbb{R}, \alpha \in \mathbb{R} \right\} \text{ alone.}$$

Thus since $\mathbb{R}^2$ has much more than 2 subspaces, $\mathbb{R}^n$ ($n > 2$) must contain much more than n subspaces.

$$\begin{aligned} 5) \quad & 3x_1 + x_2 + x_3 = b_1 \\ & x_1 - x_2 + 2x_3 = b_2 \\ & 2x_1 - 2x_2 + 4x_3 = b_3 \end{aligned}$$

Let us represent this system of equations ($AX=B$) as an augmented matrix $[A|B]$

$$\Rightarrow \left[\begin{array}{ccc|c} 3 & 1 & 1 & b_1 \\ 1 & -1 & 2 & b_2 \\ 2 & -2 & 4 & b_3 \end{array} \right]$$

on calculation, we find $\det(A) = 0$

Let us perform row operations to bring it to row echelon form

$$\left[\begin{array}{ccc|c} 3 & 1 & 1 & b_1 \\ 1 & -1 & 2 & b_2 \\ 0 & 0 & 0 & b_3 - 2b_2 \end{array} \right]$$

$$R_3 \rightarrow R_3 - 2R_2$$

At this point we find that the $\text{rank}(A) \neq 3$ as this form contains a zero vector

For consistency to be maintained

$$\text{rank}(A) = \text{rank}(A|B)$$

then rank of A must be equal to the rank of the augmented matrix.

This is only possible when the entire row of the augmented matrix

consists of 0s.

Thus $b_3 - 2b_2 = 0 \Rightarrow b_3 = 2b_2$

6) For $x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$, $y = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix}$, $x \oplus y = \begin{pmatrix} x_1 + y_1 + 5 \\ x_2 + y_2 - 7 \\ x_3 + y_3 + 1 \end{pmatrix}$

and $k \odot x = \begin{pmatrix} kx_1 + 5(k-1) \\ kx_2 - 7(k-1) \\ kx_3 + k - 1 \end{pmatrix}$

For the vector space $\mathcal{V}(\oplus, \odot)$, let the zero vector be $0 = \begin{pmatrix} 0_1 \\ 0_2 \\ 0_3 \end{pmatrix}$, $0 \in \mathcal{V}$

$\Rightarrow 0$ must act as the additive identity such that for $x \in \mathcal{V}$, $x \oplus 0 = x$

$$\Rightarrow x \oplus 0 = \begin{pmatrix} x_1 + 0_1 + 5 \\ x_2 + 0_2 - 7 \\ x_3 + 0_3 + 1 \end{pmatrix} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

$$\Rightarrow x_1 + 0_1 + 5 = x_1 \quad 0$$

$$0_1 = -5$$

$$x_2 + 0_2 - 7 = x_2 \quad 0$$

$$0_2 = 7$$

$$x_3 + 0_3 + 1 = x_3 \quad 0$$

$$0_3 = -1$$

Therefore the zero vector $\vec{0}$ must be $\begin{pmatrix} -5 \\ 7 \\ -1 \end{pmatrix}$

11

For verification,

$k \odot \vec{0} = \vec{0}$ (zero vector), for any k (scalar) $\in \mathbb{R}$

$$\Rightarrow k \odot \vec{0} = \begin{pmatrix} k(-5) + 5(k-1) \\ k(7) - 7(k-1) \\ k(-1) + k - 1 \end{pmatrix} = \begin{pmatrix} -5k + 5k - 5 \\ 7k - 7k + 7 \\ -k + k - 1 \end{pmatrix} = \begin{pmatrix} -5 \\ 7 \\ -1 \end{pmatrix} = \vec{0}$$

Hence, it is confirmed that the zero vector for this space is $\begin{pmatrix} -5 \\ 7 \\ -1 \end{pmatrix}$

2) Given

$$\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$$

considering a homogeneous system,

$$\Rightarrow \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \begin{matrix} a_{11}x_1 + a_{12}x_2 = 0 \\ a_{21}x_1 + a_{22}x_2 = 0 \end{matrix} \Rightarrow \left[\begin{array}{cc|c} a_{11} & a_{12} & 0 \\ a_{21} & a_{22} & 0 \end{array} \right]$$

If $\det(A) \neq 0$, then $\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \rightarrow$ can't be accepted

If $\det(A) = 0 \Rightarrow a_{11}x_1 + a_{12}x_2 = 0$

$$\Rightarrow a_{11}x_1 = -a_{12}x_2$$

$$\Rightarrow x_2 = -\frac{a_{11}}{a_{12}}x_1$$

let $x_1 = k \Rightarrow x_2 = -\frac{a_{11}k}{a_{12}}$

Assuming
 $a_{12} \neq 0$ and
 $a_{11} \neq 0$

then we can write $x_h = \begin{pmatrix} k \\ -\frac{a_{11}}{a_{12}} k \end{pmatrix}$

12

where $k \neq 0$, else
it will form a
zero vector solution.

Thus for any $k \neq 0$, $k \in \mathbb{R}$, x_h cannot be
a zero vector.

Now if $b \neq$ zero vector

$$\Rightarrow \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} \quad \text{where both } b_1 \neq 0 \text{ and } b_2 \neq 0$$

Let consider $x = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ then for $AX = \begin{pmatrix} a_{11}(0) + a_{12}(0) \\ a_{21}(0) + a_{22}(0) \end{pmatrix}$
 $= \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ but $B \neq 0$ in this case

Thus $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$ cannot be a particular solution
 x_p for any $B \neq 0$ vector. Good

Thus since both x_p and x_h cannot
be zero vectors, the set of ~~all~~ solutions

$\{x_{\text{complete}} = x_p + \alpha x_h\}$ does not contain
a zero vector. Hence x_{complete} is NOT
a vector subspace.